System Specification – Version 2

Akshit Dutta

Elias Hetland

Jan Adam Rozmund

Jon Erik Klakken

Martin Brekke Nilsen

Mats Anonsen

Rahul Uthayakumar

Group: 09

TTM4115 Design av kommuniserende systemer

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System Spec Version 2

Background

Online shopping and on-demand services have experienced rapid growth in recent years, placing increasing pressure on delivery companies to operate more quickly and reliably. Customers now expect short delivery times, even during busy periods. However, traditional delivery methods, particularly those that rely on cars, often struggle with traffic congestion, unexpected delays, and limited route flexibility. These challenges are especially evident in urban areas, where heavy traffic can significantly slow down deliveries.

For businesses such as restaurants and grocery stores, these delays can directly affect the quality of what they deliver. Food may arrive cold, delivery times can vary widely, and customers may end up dissatisfied. Many services also depend on human couriers, whose availability changes throughout the day and often does not match peak demand. This creates higher operational costs and makes it difficult for businesses to consistently provide fast and reliable service.

As demand continues to grow, traditional delivery methods face increasing challenges in meeting expectations for speed, reliability, and cost effectiveness. These limitations highlight the need for new and innovative approaches that can handle time-sensitive deliveries more efficiently, while reducing dependency on traffic conditions and fluctuating workforce availability

Vision

Our vision is to develop a responsive and reliable delivery platform that demonstrates how autonomous aerial delivery can overcome the limitations of traditional road-based logistics. The system aims to provide faster and more predictable delivery times by avoiding traffic congestion, reducing dependency on human couriers, and ensuring that time sensitive items reach customers when they need them.

To support this, the platform will integrate real time monitoring, automated decision making, and clear communication with users, allowing them to follow their delivery from dispatch to arrival. By combining efficient routing, transparent status updates, and consistent delivery performance, the system will showcase a modern and innovative approach to handling high demand, time-critical deliveries.

Objectives

1. The system shall enable users to request a delivery and receive clear confirmation.

   Verification: A user submits a delivery request through the interface and receives a confirmation message.

2. Provide a predictable and observable simulation of drone delivery from dispatch to arrival.

   Verification: During a test delivery, the drone's state (dispatched, in route, delivered) is shown in real time.

3. Inform users about key delivery events through timely notifications.

   Verification: Notifications appear in the user interface during a simulated delivery scenario.

4. Automatically assign a suitable drone based on distance, battery level, and package weight.

   Verification: When multiple drones are available, the system selects the drone that best matches the assignment criteria.

5. Ensure drones return to base and recharge when battery levels fall below a defined threshold.

   Verification: In simulation, a drone with a low battery triggers a return to base action and enters a charging state.

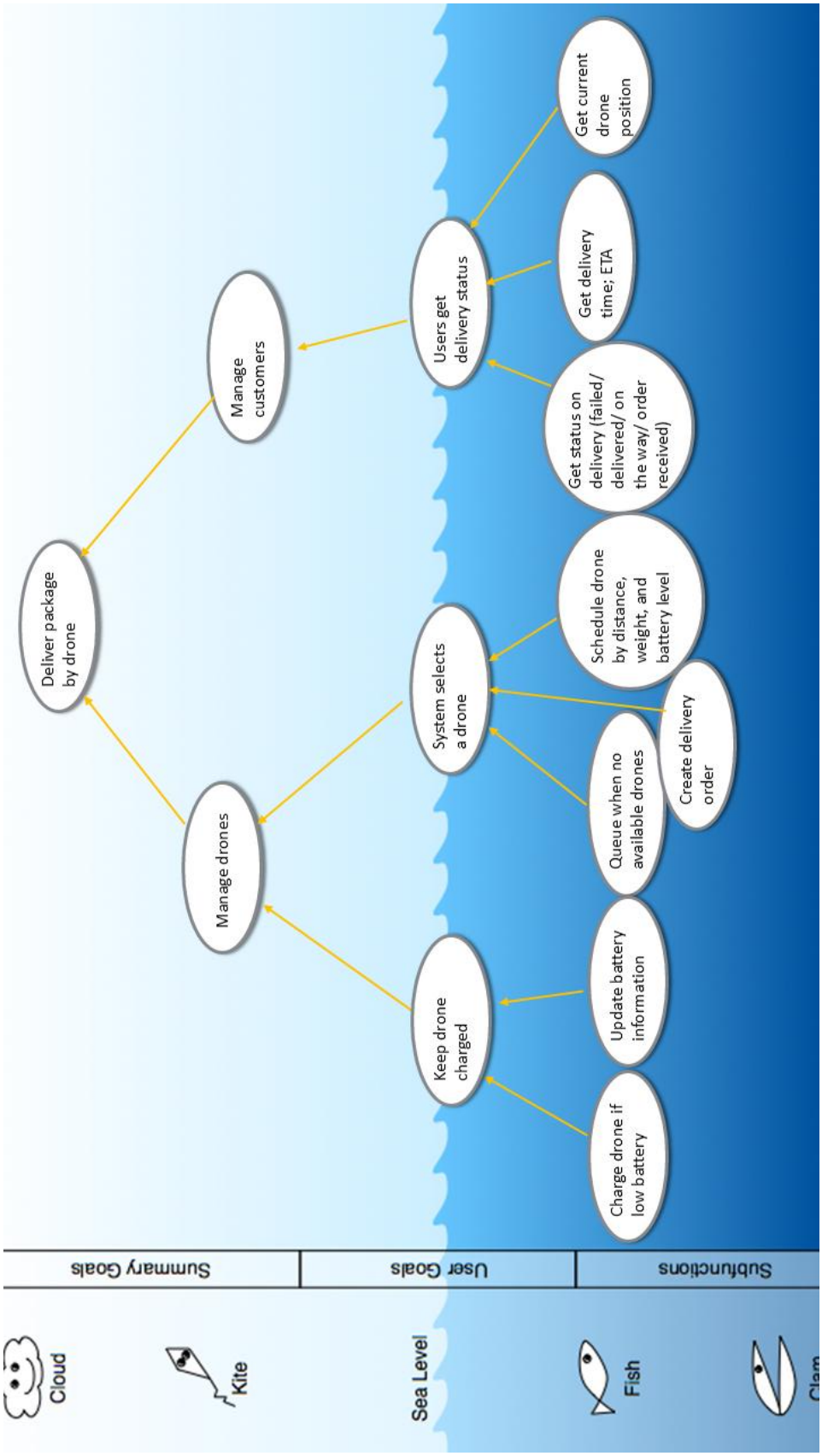
Stakeholders

Stakeholder	Major value	Attitudes	Major interests	Constraints
Managers	Improve productivity and efficiency.	Strong commitment	Cost, emloyee perscistence. Expanding the company.	Investment money
Customer	Time saving, time efficient	Strong enthusiasm with the development of the system, as it promises to significantly reduce shipping times.	Effortless user experience, dependable delivery and shorter shipping durations	Verification of the package being delivered.
Engineer staff	Enhancing systems that face challenges during transit not only boosts customer satisfaction but also elevates corporate success.	Strong commitment since the system is very dependent about it being fixed so the service goes smoothly	Job preservation	Training on company policies and security is essential. It is a space where the system can be repaired.
Fedex/ UPS/ Amazon/ DHL/ Posten/ Bring	Save money, improve efficiency	Strong commitment	Earn/save money and time with the new solution	Investment money

- It is important for stakeholders that the drone delivers the package and has a function to confirm that the package was delivered in case of fraud.

- It is crucial for stakeholders to report the number of packages sent daily, monthly, and yearly to demonstrate progress and potentially secure further investment in the system.

Use Cases


Cloud
Kite
Sea Level
Fish
Clam
Summary Goals
User Goals
Subfunctions
Deliver package by drone
Manage customers
Manage drones
Users get delivery status
System selects a drone
Keep drone charged
Get current drone position
Get delivery time; ETA
Get status on delivery (failed/ delivered/ on the way/ order received)
Schedule drone by distance, weight, and battery level
Create delivery order
Queue when no available drones
Update battery information
Charge drone if low battery

UC-1 Keep drones charged

ID and Name:	UC-1 Keep drones charged		
Created by:	Group 09	Date created:	16.03.2026
Primary Actor:	Drone operators	Secondary Actors:	Managers
Description:	The system makes sure all the drones that are not in use or have too low battery level to take a new task get charged. The system notifies drone operators when charging is needed.		
Trigger:	The battery level in a drone is too low.		
Preconditions:	PRE-1. The drone is at the base station. PRE-2. Only the authorised drone operators handle the charging. PRE-3. The system works properly.		
Postconditions:	POST-1. The battery gets charged until full.		
Normal flow:	**1.0 The drone operators charge the drone.** 1. The drone operators get a notification about low battery level. 2. They identify the correct drone and move it to the charging station. 3. They make sure the charging has properly started.		
Alternative Flows:	**1.1 No charger available.** 1. The drone operators place the drone in a queue. 2. When one of the chargers becomes available, the drone operators ensure charging the drones from the queue immediately. 3. They make sure the charging has properly started.		
Exceptions:	**1.1.E No notification received.** 1. The drone operators check the battery level displayed at the drones when the system fails. 2. They decide if the battery needs charging based on the guidelines. 3. If the battery needs charging, they move the drone to the charging station. 4. They make sure the charging has properly started.		
Priority:	High		
Frequence of use:	Each time a drone needs charging, i.e. once per 1-3 deliveries, based on how demanding the deliveries are.		
Business Rules:	n/a		
Other Information:	–		
Assumptions:	The sensor shows real battery level.		

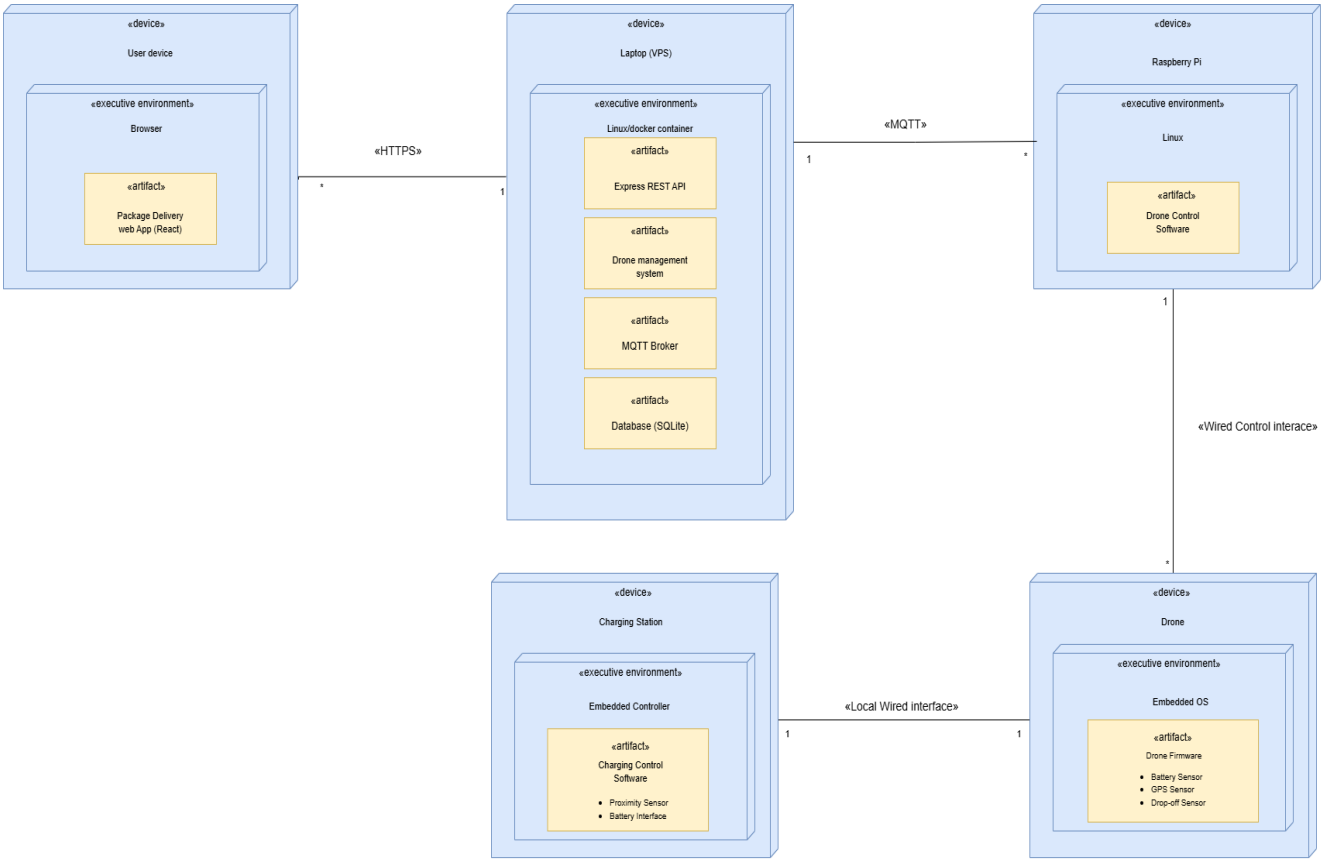
UC-2 System selects a drone

ID and Name:	UC-2 System selects a drone		
Created by:	Group 09	Date created:	16.03.2026
Primary Actor:	Collaborators	Secondary Actors:	Managers
Description:	The system selects a proper drone based on the delivery distance, package weight and current battery level. If no drone is available, it queues the requests.		
Trigger:	New order is placed in the system.		
Preconditions:	PRE-1. New order is received from collaborators (Posten etc). PRE-2. The system works properly.		
Postconditions:	POST-1. The authorised drone operators attach the package to a correct drone. POST-2. The package is sent.		
Normal flow:	**2.0 The system chooses an appropriate drone.** 1. An order is placed in the system. 2. The system chooses an appropriate drone based on the delivery distance, package weight and battery level. 3. The system notifies the drone operators about the chosen drone. 4. They identify the drone and prepare for the dispatch.		
Alternative Flows:	**2.1 No appropriate drone available.** 1. An order is placed in the system. 2. The system chooses an appropriate drone based on the delivery distance, package weight and battery level. 3. Since no appropriate drone is available, the system queues the request and sends a notification to the drone operators. 4. The system checks the queued requests each time a new drone becomes available. 5. The system notifies the drone operators about the chosen drone. 6. They identify the drone and prepare for the dispatch.		
Exceptions:	**2.1.E Invalid delivery** 1. An order is placed in the system. 2. The system detects an invalid order (no drone with full battery can manage to deliver the package). 3. The system sends a notification to the collaborator that the order does not meet the requirements.		
Priority:	High		
Frequence of use:	Each time an order is placed, approximately 100 times a day.		
Business Rules:	n/a		
Other Information:	–		
Assumptions:	The orders are received in a valid form.		

UC-3 User gets delivery status

ID and Name:	UC-3 Users get delivery status		
Created by:	Group 09	Date created:	16.03.2026
Primary Actor:	Users	Secondary Actors:	Collaborators, Managers
Description:	The user is notified when the order is placed, the package is on the way, has arrived or failed. The user sees the current drone position and ETA.		
Trigger:	New delivery status is registered.		
Preconditions:	PRE-1. The user PRE-2.		
Postconditions:	POST-1. The user has received all the notifications at the right time.		
Normal flow:	**3.0 User gets notified.** 1. A new status is registered in the system. 2. The system sends a proper notification to the customer and/or the drone position is updated.		
Alternative Flows:	–		
Exceptions:	**3.1.E The system that manages notifications failed.** 1. A new status is registered in the system. 2. The managers send a notification about the delivered package manually.		
Priority:	Middle		
Frequence of use:	Each time a new status is registered (usually 3 per delivery + real time drone position).		
Business Rules:	n/a		
Other Information:	–		
Assumptions:	The user has access to the internet.		

Deployment Diagram

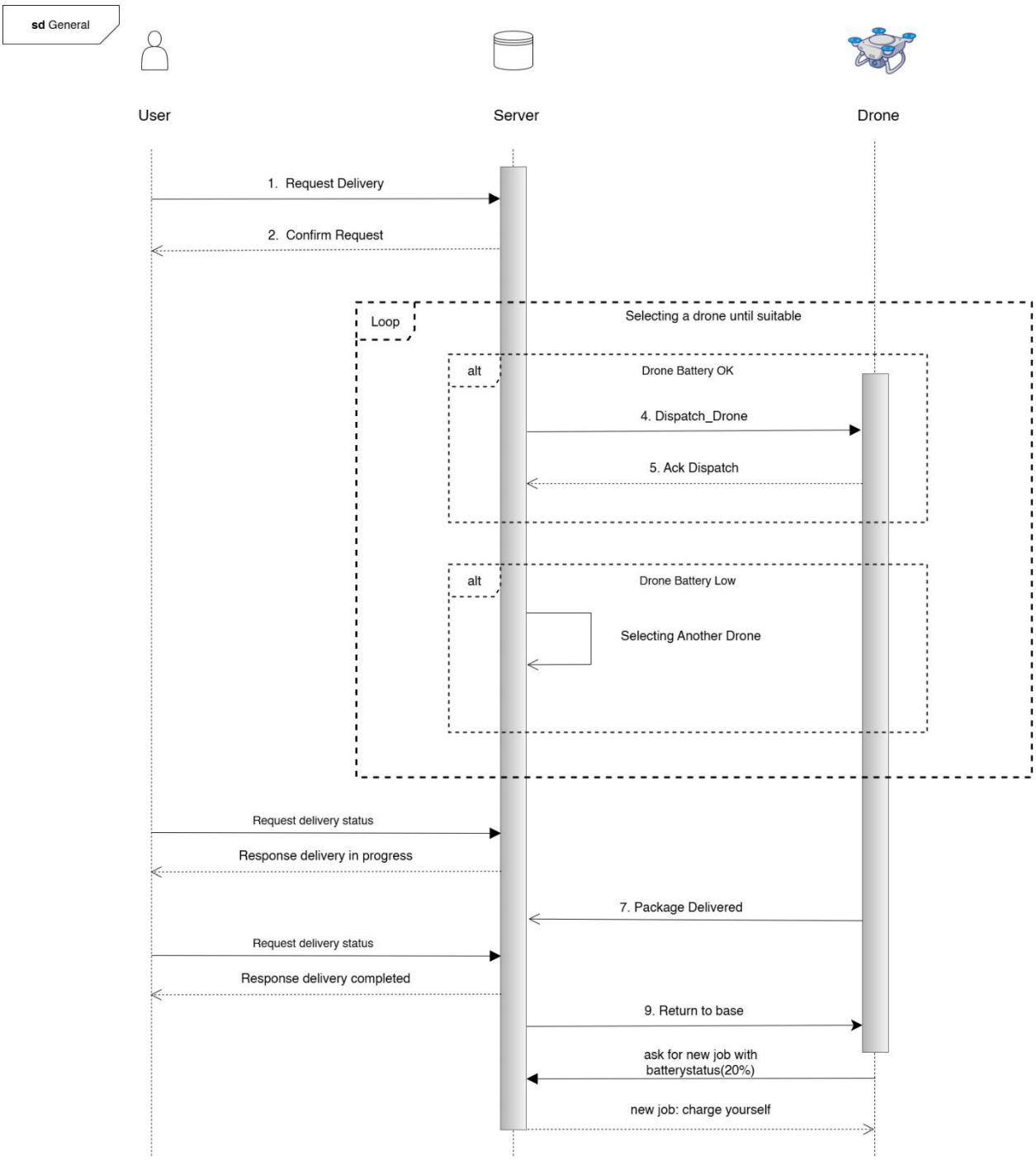


Explanation

This diagram shows how the charging system works from the user's device all the way down to the hardware. The user interacts with a simple app on their phone or computer, which sends requests to the cloud backend using HTTPS. The cloud processes these requests and communicates with the Raspberry Pi through MQTT. The Raspberry Pi acts as the local controller and sends wired control signals to the charging meter. The charging meter runs on an embedded controller and manages the actual charging process using components like the remote switch and power sensor. Finally, the device receives local wired signals from the charging meter to start or stop charging.

Sequence Diagrams

General drone sequence diagram

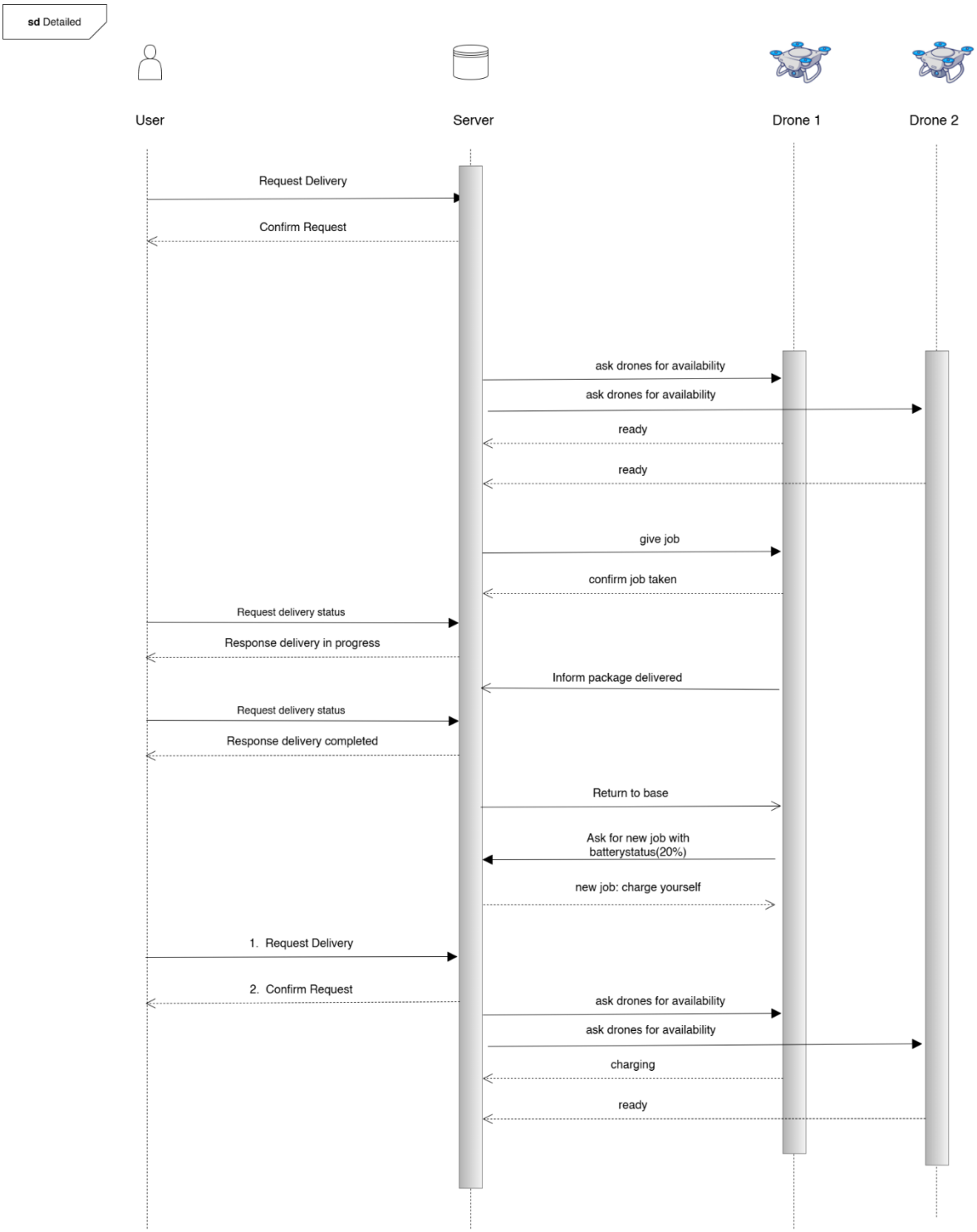


Explanation for the general sequence diagram

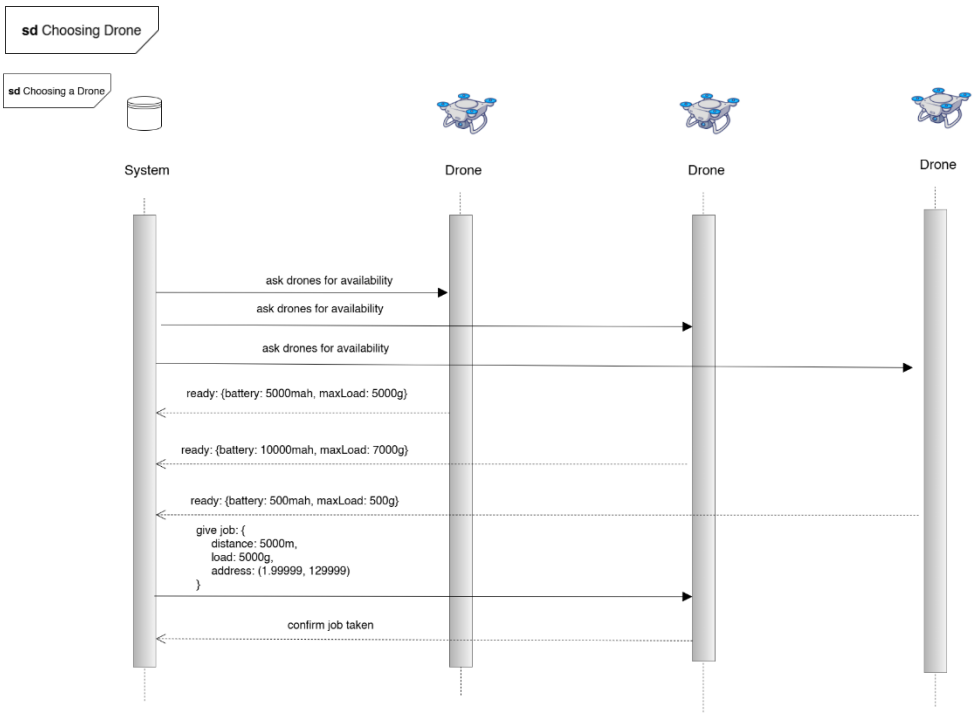
This sequence diagram shows how the drone delivery process works from start to finish. First, the User sends a delivery request to the system, which confirms it. The system then looks for a suitable drone by checking each drone's battery level. If a drone has low battery, the system skips it and tries another one. When a drone with enough battery is found, the system dispatches it and receives an acknowledgment.

After the drone is sent out, the system updates the User that the delivery is in progress. Once the package is delivered, the system sends another update confirming that the delivery is completed. Finally, the drone returns to base and begins charging.

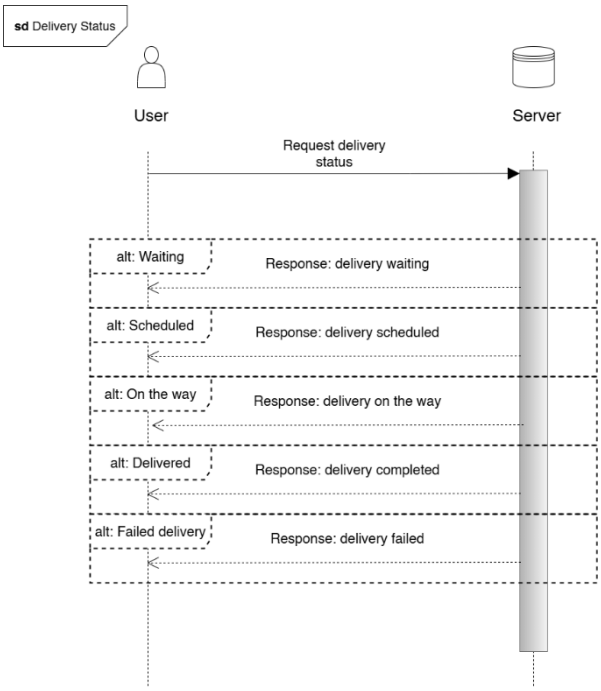
Sequence diagram in detail:



Choosing a Drone

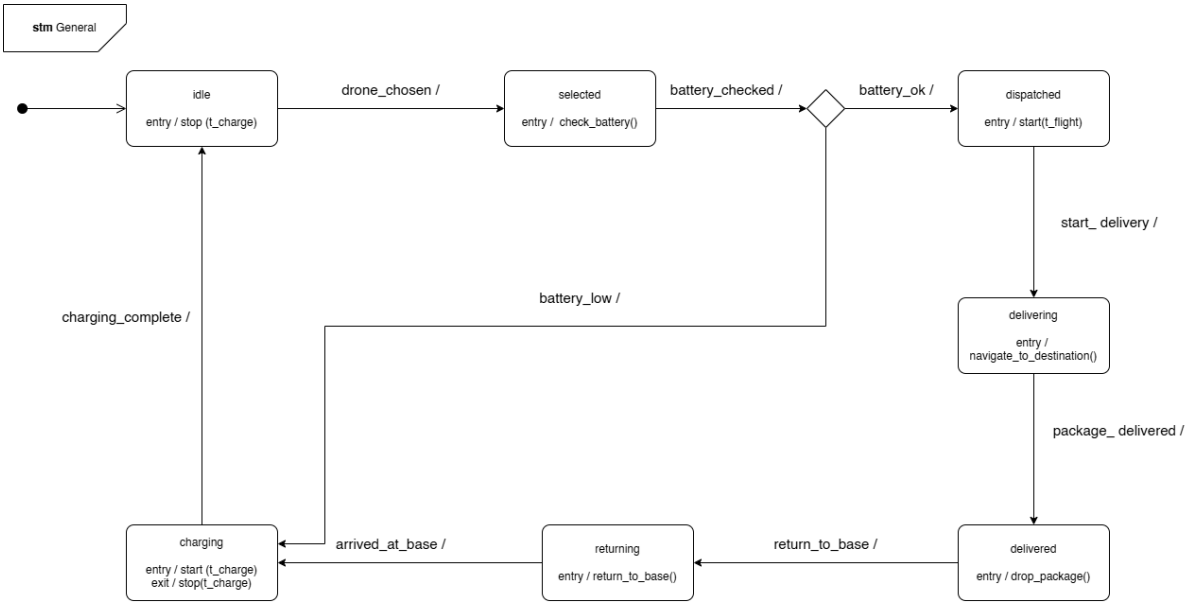


Delivery Status



State Machines

General drone State Machine

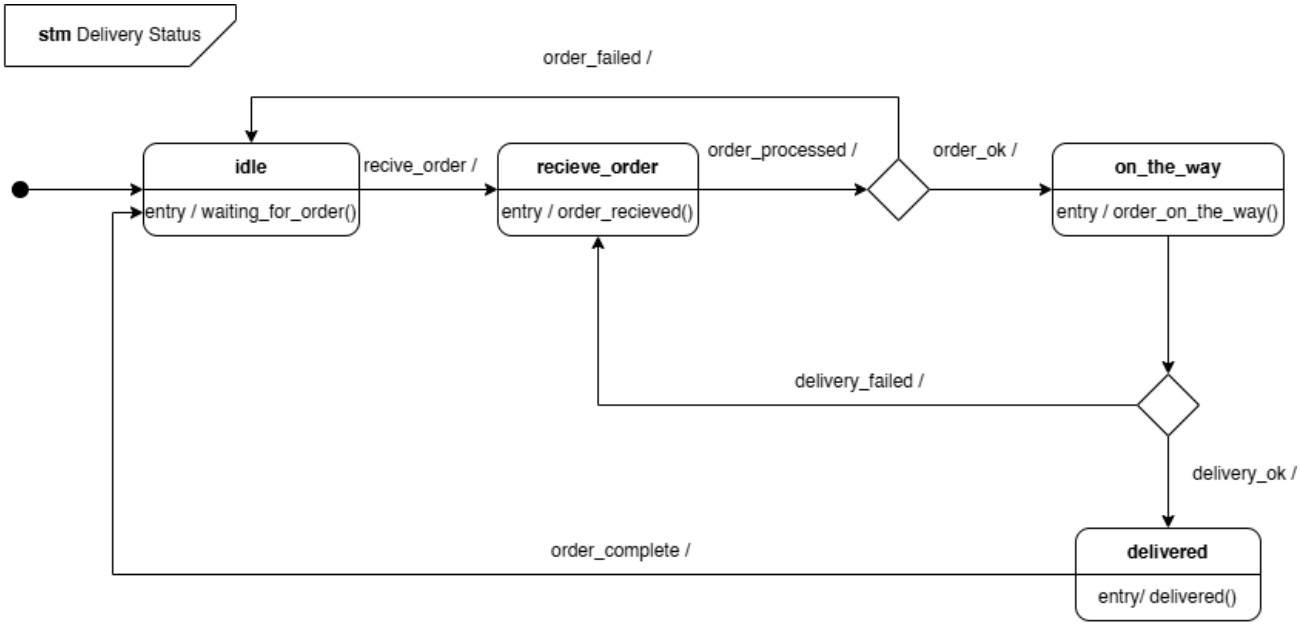


Explanation of the general drone state machine

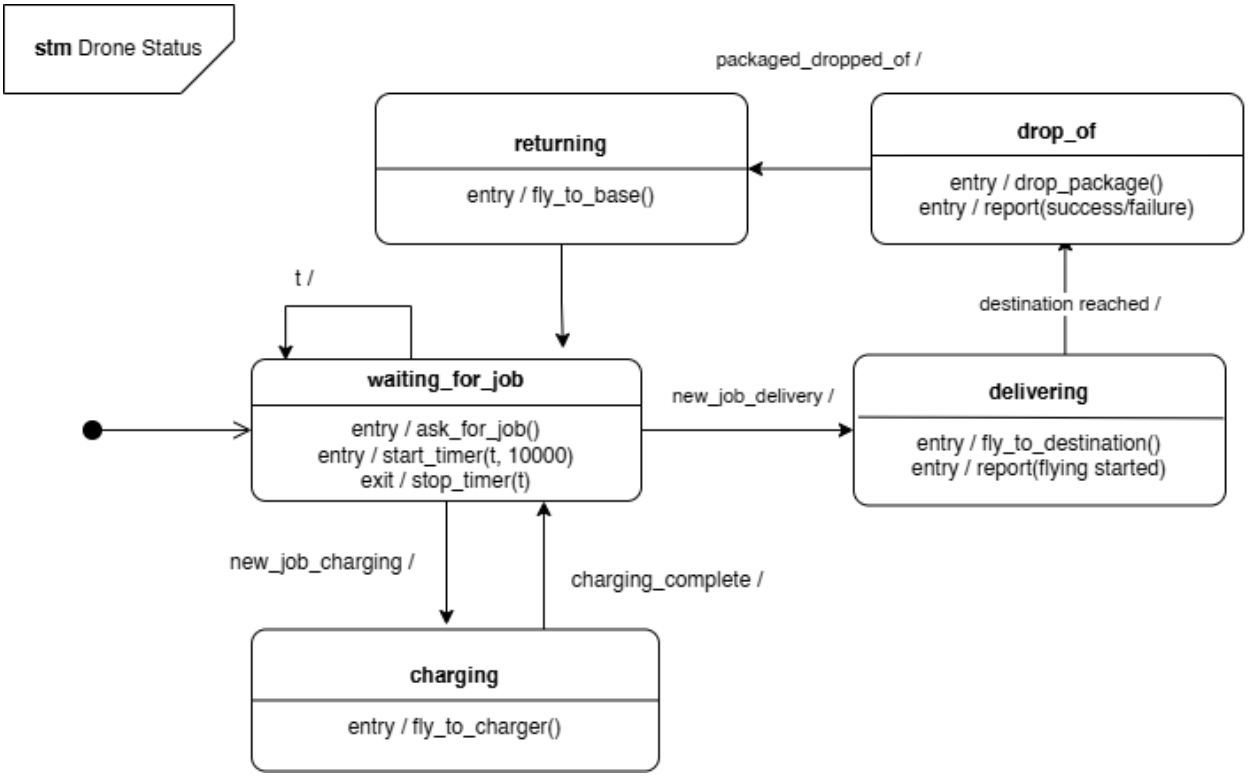
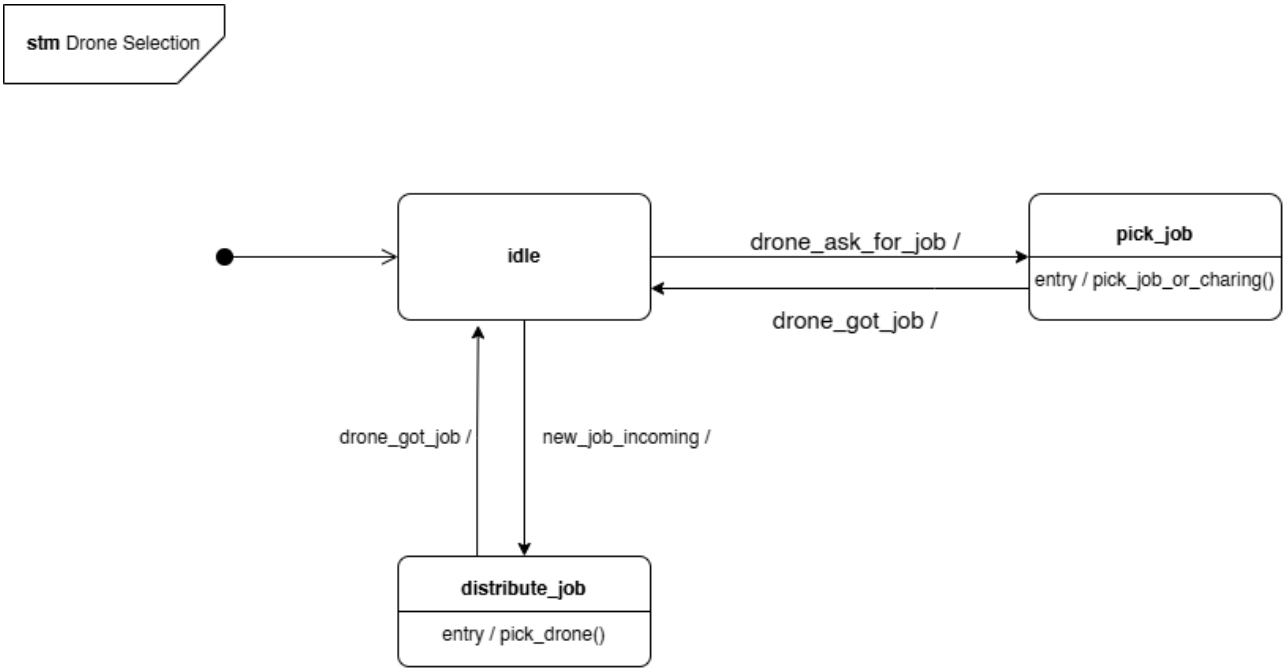
This state machine shows the different stages a drone goes through during a delivery. The drone starts in the *idle* state, waiting for a task. When the system chooses it for a delivery, it moves to *selected* and checks its battery. A decision is then made: if the battery level is good, the drone is dispatched; if the battery is low, it goes back to charging before it can be used again.

Once dispatched, the drone begins the delivery process. It travels to the destination (*delivering*), drops off the package (*delivered*), and then returns to base (*returning*). When it arrives, it enters the *charging* state until the battery is full. After charging is complete, the drone returns to *idle* and is ready for the next delivery.

Delivery status



Drone Status



Drone Selection